Fig. 9.1 Standard wire gauge plate

of the conductor to be found is inserted into the notches. The wire which just fits with a particular notch, stated to be the exact size of the wire. The number represented near that notch will be the gauge number of that wire.

By finding gauge number of a wire, we can determine the amount of electric current the wire can safely carry, as well as its electrical resistance and weight per unit length.

We can measure the diameter of the winding wire by using the instrument micrometer (screw gauge). After measuring the diameter of the wire, using the above instrument, we can find the gauge number of the wire from the following table given below.

S.W.G No	Standard wire gauge SWG			SWG No	Standard wire gauge SWG		
	Inch	mm	Area mm²		Inch	mm	Area mm²
0	0.324	8.23	53.17	26	0.018	0.457	0.164
1	0.3	7.62	45.6	27	0.016	0.417	0.136
2	0.276	7.01	38.6	28	0.015	0.376	0.111
3	0.252	6.401	32.2	29	0.014	0.345	0.0937
4	0.232	5.893	27.3	30	0.012	0.315	0.0779
5	0.212	5.385	22.8	31	0.012	0.295	0.0682
6	0.192	4.877	18.7	32	0.011	0.274	0.0591
7	0.176	4.47	15.7	33	0.01	0.254	0.0507
8	0.16	4.064	13	34	0.009	0.234	0.0429
9	0.144	3.658	10.6	35	0.008	0.213	0.0357
10	0.128	3.251	8.3	36	0.008	0.193	0.0293
11	0.116	2.946	6.82	37	0.007	0.173	0.0234
12	0.104	2.642	5.48	38	0.006	0.152	0.0182
13	0.092	2.337	4.29	39	0.005	0.132	0.0137
14	0.08	2.032	3.24	40	0.005	0.122	0.0117
15	0.072	1.829	2.63	41	0.004	0.112	0.0098
16	0.0064	1.626	2.07	42	0.004	0.102	0.0082
17	0.056	1.422	1.59	43	0.004	0.091	0.0065
18	0.048	1.219	1.17	44	0.003	0.081	0.0636
19	0.04	1.016	0.811	45	0.003	0.071	0.0557
20	0.036	0.914	0.657	46	0.002	0.061	0.0479
21	0.032	0.813	0.519	47	0.002	0.051	0.04
22	0.028	0.711	0.397	48	0.002	0.041	0.0322
23	0.024	0.61	0.292	49	0.001	0.03	0.0236
24	0.022	0.559	0.245	50	0.001	0.025	0.0196
25	0.02	0.508	0.203	BASIS: IS 13730-0-1/IEC 317-0-1			

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